Quiz for Electrodynamics

1 Single Choice Questions

1. Which one implies the existence of a vector magnetic potential $\mathbf{A}$ (B) ?

   A. $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$

   B. $\nabla \cdot \mathbf{B} = 0$

   C. $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

   D. $\nabla \times \mathbf{B} = \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}_e$

2. Which one implies the existence of a gauge field ϕ? (A)

   A. $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$

   B. $\nabla \cdot \mathbf{B} = 0$

   C. $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

   D. $\nabla \times \mathbf{B} = \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}_e$

3. Which one about gauge choices is **NOT true**? (D)

   A. The Lorentz gauge is given by $\nabla \cdot \mathbf{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0$

   B. The Coulomb gauge is given by $\nabla \cdot \mathbf{A} = 0$

   C. In the Lorentz gauge, the Maxwell equations can be rewritten as:

   $$\nabla^2 \mathbf{A} - \frac{1}{c^2}\frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}, \quad \nabla^2 \phi - \frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} = -\frac{\rho}{\epsilon_0}$$

   D. Gauge is a constraint that arises from the requirements of theory.

4. Which of the following statements about the vector potential $\mathbf{A}$ and the scalar potential ϕ is **NOT true**? (A)

A. $\mathbf{A}$ and ϕ are mathematical constructs introduced artificially, but both have been experimentally proven to exist.

B. $\mathbf{A}$ and ϕ are functions of spacetime, i.e., $\mathbf{A} = \mathbf{A}(x, y, z, t)$, $\phi = \phi(x, y, z, t)$.

C. Once $\mathbf{A}$ and ϕ are determined, they cannot be altered.

D. Once $\mathbf{A}$ and ϕ are known, the electric and magnetic fields are determined via $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}$.

5. Which one about electric fields is **NOT true**? (C)

A. The electric field exists at every point in spacetime, and can thus be written as $\mathbf{E} = \mathbf{E}(x, y, z, t)$.

B. The curl of the electric field is always zero.

C. The curl of the electric field can be written as $\nabla \times \mathbf{E}$, and its direction is opposite to that of $\mathbf{B}$.

D. The divergence of the electric field is a scalar.

6. Which one about electric fields is **NOT true**? (C)

A. Both electric and magnetic fields are functions of spacetime.

B. The electric field can be written as $\mathbf{E} = E_x(x, y, z, t)\hat{i} + E_y(x, y, z, t)\hat{j} + E_z(x, y, z, t)\hat{k}$.

C. The electric field can be written as $\mathbf{E} = E_x(x, t)\hat{i} + E_y(y, t)\hat{j} + E_z(z, t)\hat{k}$.

D. The curl of the electric field can also be a function of (x, y, z, t).

7. Which of the following statements about electromagnetic fields is **NOT true**? D

A. Electromagnetic waves are formed by the interaction of electric and magnetic fields.

B. Electromagnetic waves are formed by the transformation of charges and currents, and once formed, they propagate in a vacuum or medium, independent of the charges and currents.

C. Plane electromagnetic waves are a specific form of electromagnetic waves.

D. Electromagnetic waves are formed by the transformation of charges and currents, and their propagation requires continuous transformation of charges and currents.

8. Which of the following statements about the propagation of electromagnetic waves is **NOT true**? (A)

   A. The direction of propagation of a plane electromagnetic wave can be written as $\mathbf{k} = k_x(x, y, z, t)\hat{i} + k_y(x, y, z, t)\hat{j} + k_z(x, y, z, t)\hat{k}$.

   B. The direction of propagation of a plane electromagnetic wave is perpendicular to both the electric field and magnetic field, which can be expressed mathematically as $\mathbf{k} \cdot \mathbf{E} = 0 = \mathbf{k} \cdot \mathbf{B}$.

   C. The direction of propagation of a plane electromagnetic wave is perpendicular to both the electric field and magnetic field, and this can be derived from the fact that the divergence of both the electric field and magnetic field is zero.

   D. The direction of propagation of a plane electromagnetic wave can be written as $\mathbf{k} = k_x(x, t)\hat{i} + k_y(y, t)\hat{j} + k_z(z, t)\hat{k}$.

9. Which of the following statements about planar electromagnetic waves is **NOT true**? (D)

   A. Once the electric component $\mathbf{E}$ of the plane electromagnetic wave is known, the magnetic component $\mathbf{B}$ is also determined.

   B. For electromagnetic waves, the electric and magnetic fields can be described by Maxwell's equations: $\nabla \cdot \mathbf{E} = 0, \nabla \cdot \mathbf{B} = 0, \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \nabla \times \mathbf{B} = \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t}$, but knowing the electric field alone does not automatically determine the magnetic field.

   C. The interaction between the electric and magnetic fields can be observed from the equations $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and $\nabla \times \mathbf{B} = \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t}$.

   D. Once the electromagnetic wave is specified, the values of $\mathbf{E}_0$, $\mathbf{k}$, and ω are fixed.

10. Which of the following statements about electromagnetic wave propagation in matter is **NOT true**? (A)

    A. Like in a vacuum, electromagnetic waves of different frequencies travel at the same speed in matter. 色散

    B. Like in a vacuum, electromagnetic waves of different frequencies have a fixed speed in matter.

    C. Electromagnetic waves in matter gradually attenuate.

    D. In matter, the wave number k of electromagnetic waves is complex.

11. Regarding the wave equation for electromagnetic waves in a cavity, which of the following statements is **NOT true**? (D)

    A. The electromagnetic waves in the cavity are standing waves, with both the electric and magnetic fields at each point having definite magnitudes.

    B. The form of the electromagnetic wave in the cavity is determined by the boundary conditions.

    C. Based on the frequency $\omega_{mnp} = \frac{\pi}{\sqrt{\mu\epsilon}}\sqrt{\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2 + \left(\frac{p}{L_3}\right)^2}$, we see that once the dimensions of the cavity are fixed, only certain frequencies of electromagnetic waves can exist in the cavity.

    D. The electromagnetic wave in the cavity $\mathbf{E}_x = A_1 \cos(k_x x)\sin(k_y y)\sin(k_z z)$ shows that the wave propagates along the z-axis.

12. We derive $\mathbf{A}(x,y,z,t) = \mathbf{B}(x,y,z,t)$ from the equation $\hat{e}_n \times \mathbf{A}(x,y,z,t) = \hat{e}_n \times \mathbf{B}(x,y,z,t)$. Which of the following statements regarding this derivation is correct? B

    A. If we can find two distinct vectors $\hat{e}_n \times \mathbf{A}_0 = \hat{e}_n \times \mathbf{B}_0$, we can conclude that our derivation above is incorrect.

    B. If the condition $\hat{e}_n \times \mathbf{A}(x,y,z,t) = \hat{e}_n \times \mathbf{B}(x,y,z,t)$ holds for all (x,y,z,t), then it must lead to the conclusion $\mathbf{A}(x,y,z,t) = \mathbf{B}(x,y,z,t)$.

    C. We can express $\mathbf{A}(x,y,z,t) = \mathbf{E}_0 e^{i\mathbf{k}\cdot\mathbf{r}} e^{-i\omega t} + \mathbf{E}_0' e^{i\mathbf{k}'\cdot\mathbf{r}} e^{-i\omega' t}$ and $\mathbf{B}(x,y,z,t) = \mathbf{E}_0'' e^{i\mathbf{k}''\cdot\mathbf{r}} e^{-i\omega'' t}$, which can be used to derive Snell's Law.

D. The relationship $\hat{e}_n \times \mathbf{A}(x, y, z, t) = \hat{e}_n \times \mathbf{B}(x, y, z, t)$ can only be valid for certain boundary conditions, and does not necessarily imply $\mathbf{A}(x, y, z, t) = \mathbf{B}(x, y, z, t)$.

2 Calculations

2.1 Verifying the Divergence and Curl Relations

We define the Nabla operator as:

$$\nabla = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

We are tasked with verifying the following two relations:

1. The divergence of the curl is zero:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

1. $\nabla \times X = \left(\frac{\partial X_z}{\partial y} - \frac{\partial X_y}{\partial z}, \frac{\partial X_x}{\partial z} - \frac{\partial X_z}{\partial x}, \frac{\partial X_y}{\partial x} - \frac{\partial X_x}{\partial y}\right)$

$\nabla \cdot (\nabla \times X) = \frac{\partial}{\partial x}\left(\frac{\partial X_z}{\partial y} - \frac{\partial X_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial X_x}{\partial z} - \frac{\partial X_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial X_y}{\partial x} - \frac{\partial X_x}{\partial y}\right)$

$= 0$

2. The curl of the gradient is zero:

$$\nabla \times (\nabla \phi) = 0$$

2. $\nabla \times (\nabla \phi) = \left(\frac{\partial}{\partial y}\left(\frac{\partial \phi}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial \phi}{\partial y}\right)\right)i + \left(\frac{\partial}{\partial z}\left(\frac{\partial \phi}{\partial x}\right) - \frac{\partial}{\partial x}\left(\frac{\partial \phi}{\partial z}\right)\right)j + \left(\frac{\partial}{\partial x}\left(\frac{\partial \phi}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial \phi}{\partial x}\right)\right)k$

$= 0$

2.2 Gradient of the Inverse of Distance

Calculate the gradient of the function $\frac{1}{r}$, where r is the distance between two points.

$$r = |\mathbf{r} - \mathbf{r}'|$$

$r = |\vec{r} - \vec{r}'| = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$

$\nabla\left(\frac{1}{r}\right) = \nabla\left(\frac{1}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}}\right)$

$= -\frac{\vec{r} - \vec{r}'}{r^3}$

2.3 Comparison of Mechanical and Electromagnetic Waves

Write the forms of the mechanical wave equation and electromagnetic wave equation, and then compare their characteristics.

$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$

$\frac{\partial^2 E}{\partial t^2} = c^2 \nabla^2 E$

same form

constant speed c

u - scalar

E - vector

2.4 Snell's law

In deriving Snell's Law, we start with the following expression:

$$\mathbf{E}_0 e^{i\mathbf{k}\cdot\mathbf{r}} e^{-i\omega t} + \mathbf{E}_0' e^{i\mathbf{k}'\cdot\mathbf{r}} e^{-i\omega' t} = \mathbf{E}_0'' e^{i\mathbf{k}''\cdot\mathbf{r}} e^{-i\omega'' t}$$

From this, we can infer that under appropriate boundary conditions, the spatial distribution of the electric field should satisfy the following relation:

$$\mathbf{E}_0 e^{i\mathbf{k}\cdot\mathbf{r}} + \mathbf{E}_0' e^{i\mathbf{k}'\cdot\mathbf{r}} = \mathbf{E}_0'' e^{i\mathbf{k}''\cdot\mathbf{r}}$$

This reasoning implies that under boundary conditions, the amplitude and wave vector of the electric field must satisfy specific relations, which ultimately leads to the derivation of Snell's Law.

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

2.5 Fresnel Equations

We consider the reflection and refraction of electromagnetic waves at the boundary between two media. Let the incident electromagnetic wave be confined to the x-z plane, and assume that the electric field is oriented perpendicular to the x-y plane, propagating outward. Based on the boundary conditions:

$$\hat{e}_n \times (\mathbf{E}_2 - \mathbf{E}_1) = 0 \quad \text{(electric field boundary condition)},$$

$$\hat{e}_n \times (\mathbf{H}_2 - \mathbf{H}_1) = 0 \quad \text{(magnetic field boundary condition)},$$

$E_1 = E_0 e^{ik\cdot r - i\omega t}$
$E_2 = E_0' e^{ik'\cdot r - i\omega' t}$
$E_3 = E_0'' e^{ik''\cdot r - i\omega'' t}$

Please write down the form of the electromagnetic waves, and then derive the following two equations using the boundary conditions above:

1. The total electric field on the other side of the boundary is given by:

$$\mathbf{E} + \mathbf{E}' = \mathbf{E}''.$$

2. The magnetic field components satisfy the relation:

$$\mathbf{H}\cos\theta - \mathbf{H}'\cos\theta' = \mathbf{H}''\cos\theta''.$$

2.6 Planar waves

For a plane electromagnetic wave, the electric field can be expressed as:

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)} = \mathbf{E}_0 e^{i\mathbf{k}\cdot\mathbf{r}} e^{-i\omega t},$$

Now, starting from this expression, calculate the following:

1. The divergence of the electric field:

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t)$$

1. $\nabla \cdot E(r,t) = \nabla \cdot (E_0 e^{i(k\cdot r - wt)}) = E_0 \cdot \nabla e^{i(k\cdot r - wt)}$
$= E_0 \cdot ik e^{i(k\cdot r - wt)} = i(k\cdot E_0) E(r,t)$

2. The curl of the electric field:

$$\nabla \times \mathbf{E}(\mathbf{r}, t)$$

2. $\nabla \times E(r,t) = E_0 \times \nabla e^{i(k\cdot r - wt)} = E_0 \times (ike^{i(k\cdot r - wt)})$
$= i(k \times E_0) e^{i(k\cdot r - wt)} = i(k \times E_0) E(r,t)$

3. The gradient of the exponential term $e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$:

$$\nabla e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$$

3. $\nabla e^{i(k\cdot r - wt)} = \nabla e^{i(k\cdot r)} e^{-iwt}$
$\nabla e^{i(k\cdot r)} = ike^{i(k\cdot r)}$
$\nabla e^{i(k\cdot r - wt)} = ike^{i(k\cdot r - wt)}$

4. The time derivative of the electric field:

$$\frac{\partial}{\partial t}\mathbf{E}(\mathbf{r}, t)$$

4. $\frac{\partial}{\partial t} E(r,t) = \frac{\partial}{\partial t}(E_0 e^{i(k\cdot r - wt)}) = E_0 \cdot (-iwe^{i(k\cdot r - wt)})$
$= -iw E(r,t)$

These calculations will help you explore the behavior of the electric field in the context of Maxwell's equations.